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ClimateWins – Final report

January 2026



OBJECTIVES

- ▶ 1) Identify weather patterns outside the regional norm in Europe
- ▶ 2) Determine if unusual weather patterns are increasing
- ▶ 3) Generate possibilities for future weather conditions over the next 25 to 50 years based on current trends
- ▶ 4) Determine the safest places for people to live in Europe over the next 25 to 50 years



THOUGHT EXPERIMENTS

All thought experiments (TEs) will tie in with ClimateWins' objectives:

- ▶ 1) Weather change prediction: machine learning to predict future weather condition changes over next 25-50 years
- ▶ 2) Weather image prediction: machine learning to generate image of what an average week should look like based on past data
- ▶ 3) Weather safety prediction: machine learning to predict if a day will have safe weather or not



BEST AVAILABLE MACHINE LEARNING OPTIONS

- ▶ Artificial Neural Networks (ANNs). Useful over time as scale well, handle more variables, and learn complex weather data (change in conditions) for linear regression requirements. Lower accuracy (72%) but this can be improved using optimised hyperparameters
- ▶ Generative Adversarial Networks (GANs). Required for image prediction and generation, but costly and require high computational power
- ▶ Random Forests. When using hyperparameters had excellent accuracy (2010s at 96%) for classification, to use on safe and unsafe weather
- ▶ Convoluted Neural Networks (CNNs) also worth considering alongside Random Forests to compare results (although accuracy was lower at 68%)



DATA REQUIRED

- ▶ Historic weather data set (1960s to 2020s) can be strengthened as largely missing data for important conditions snow and wind speed, and incomplete for Gdansk, Roma and Tours
- ▶ Current and complete weather data, including all above
- ▶ Image data from weather stations, historic and current. Use with Kaggle data set for training GANs
- ▶ Data on safe and unsafe days; this can simply be if a weather alert was issued on each day or not (this time correct for Sonnblick!)



TE1 – WEATHER CHANGE

Weather change prediction: machine learning to predict future weather condition changes over next 25-50 years

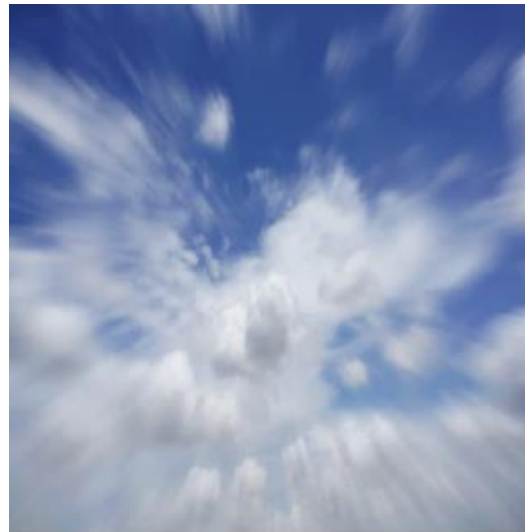
- ▶ ClimateWins wants to know if the weather is changing and what patterns are forming
- ▶ Use ANNs for linear regression predicting next 25-50 years, can plot predicted changes over time for each of the conditions in each of the weather stations
- ▶ To manage costs, and resources, early investigation could show which conditions fluctuate the most in which areas, which we could then target future analysis on



TE2 – WEATHER IMAGES

Weather image prediction: machine learning to generate image of what an average week should look like based on past data

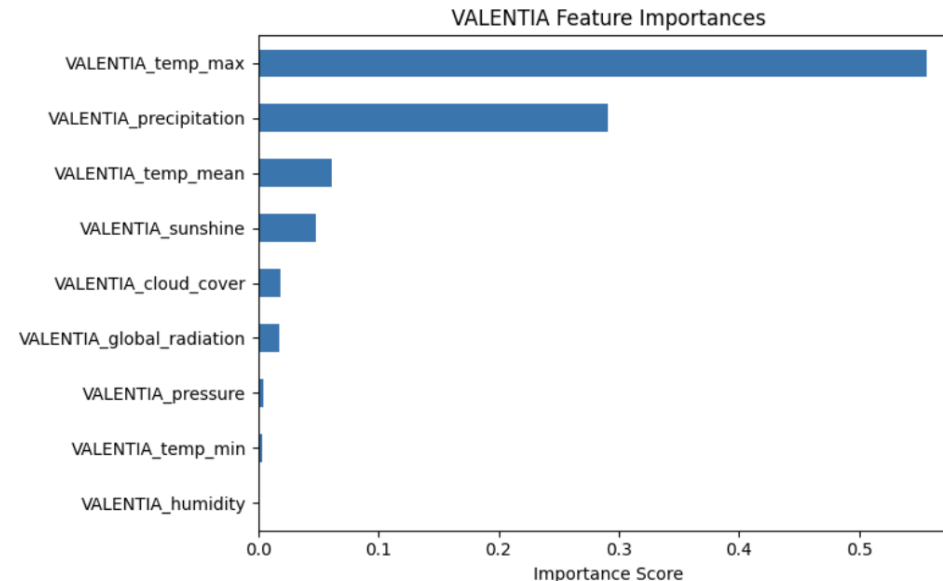
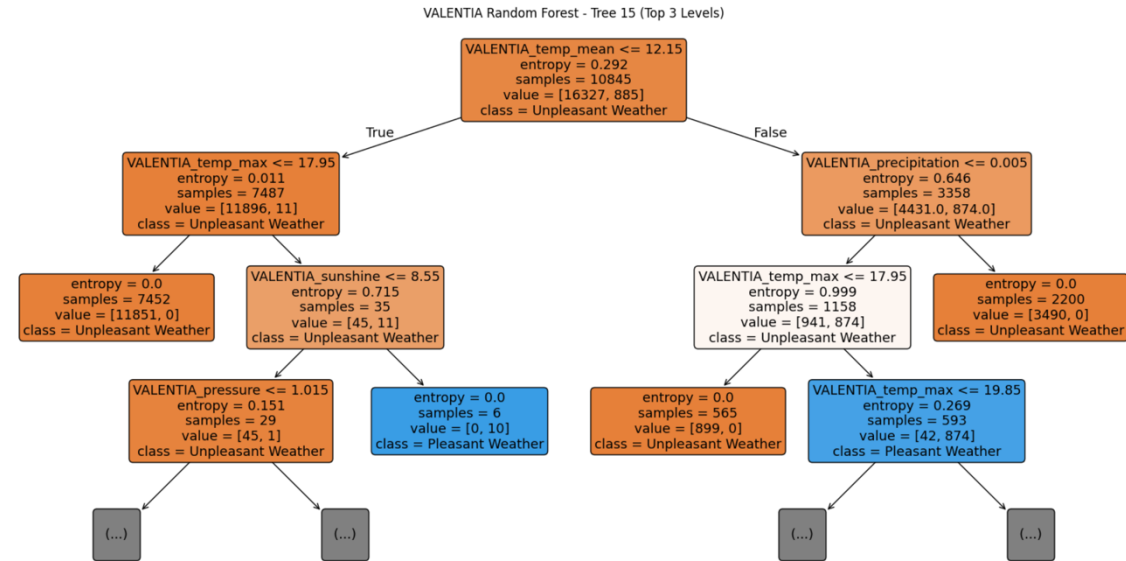
- ▶ ClimateWins wants to know future weather patterns and conditions
- ▶ GANs could learn from current images and predict the type of weather, while generating images for future conditions
- ▶ If they can predict what an average week looks like in an area, unsafe weather can be avoided or mitigated, and useful for event planning/tourism



TE3 – WEATHER SAFETY

Weather safety prediction: machine learning to predict if a day will have safe weather or not

- ▶ ClimateWins wants to know which areas will be the safest in the next 25-50 years
- ▶ Had successful results using Random Forests for comparing weather conditions on pleasant/unpleasant days – can do same with safe/unsafe days from weather alerts
- ▶ Analyzing the feature importances. Will also show us which variables contribute most to the safe/unsafe classification, which we could tie in with the results from TE1!



RECOMMENDED THOUGHT EXPERIMENT AND NEXT STEPS

I feel TE1 weather change prediction has the most potential but depends on ClimateWins' resources:

- ▶ It can answer objectives 1-3, and if our results show changes or patterns in dangerous conditions such as high wind speeds and rainfall especially in specific areas, could also answer objective 4 (safe/unsafe)
- ▶ However, must bear in mind Random Forests are generally cheaper to run (GANs most expensive!) and we are looking at large amount of historic and current data, with many conditions
- ▶ Would need to know ClimateWins' budgets, resources, and computational power to decide which approach is best, or even better, if we can be greedy and try them all!



THANK
YOU!

QUESTIONS?

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